

SMART INDIA HACKATHON 2025



- **Problem Statement ID – 25083**
- **Problem Statement Title- Digital Health Record Management System for migrant workers in Kerala aligned with sustainable development goals.**
- **Theme- HealthTech**
- **PS Category- Software**
- **Team ID- 91967**
- **Team Name – My Health**



Problem Statement:

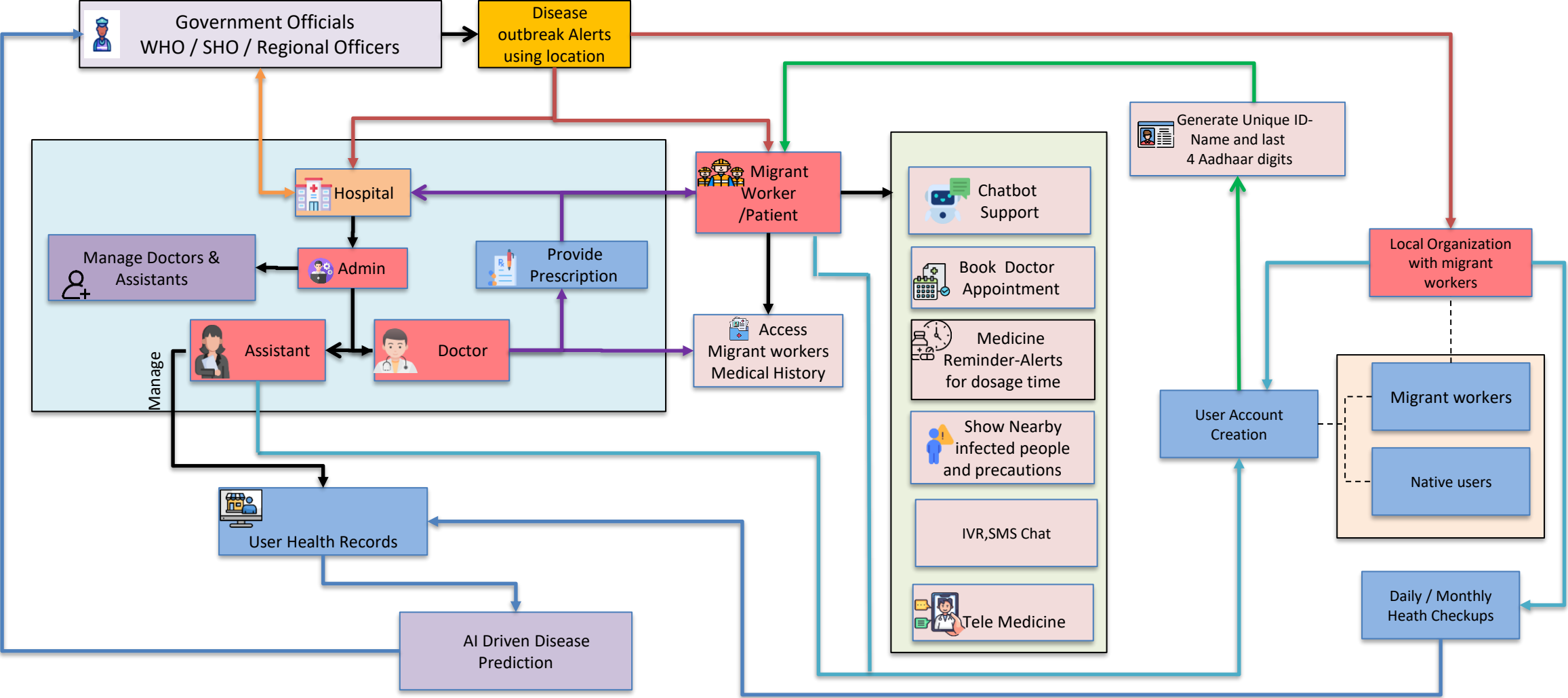
-  Kerala hosts a large **migrant population** with limited healthcare access.
-  **Lack of unified health record system** due to fragmented and inaccessible data.
-  Migrants often act as **carriers of infectious diseases**, raising **community risk**.
-  **Delayed outbreak detection** due to lack of monitoring & early warnings.
-  **Inequitable healthcare access** causes migrants to struggle with treatment & insurance.

Proposed Solution:

-  **Unified Health Identity (UHI):** One worker = one portable, secure health record across hospitals.
-  **AI-Powered Prediction & Alerts:** Early detection of diseases + proximity infection notifications.
-  **Telemedicine & Digital Prescriptions:** Remote consultations with instant, paperless records.
-  **Smartwatch & IoT Integration:** Real-time vitals, SOS alerts, and outbreak prevention.
-  **Government & Employer Dashboards:** Disease heatmaps, compliance tracking, and analytics.

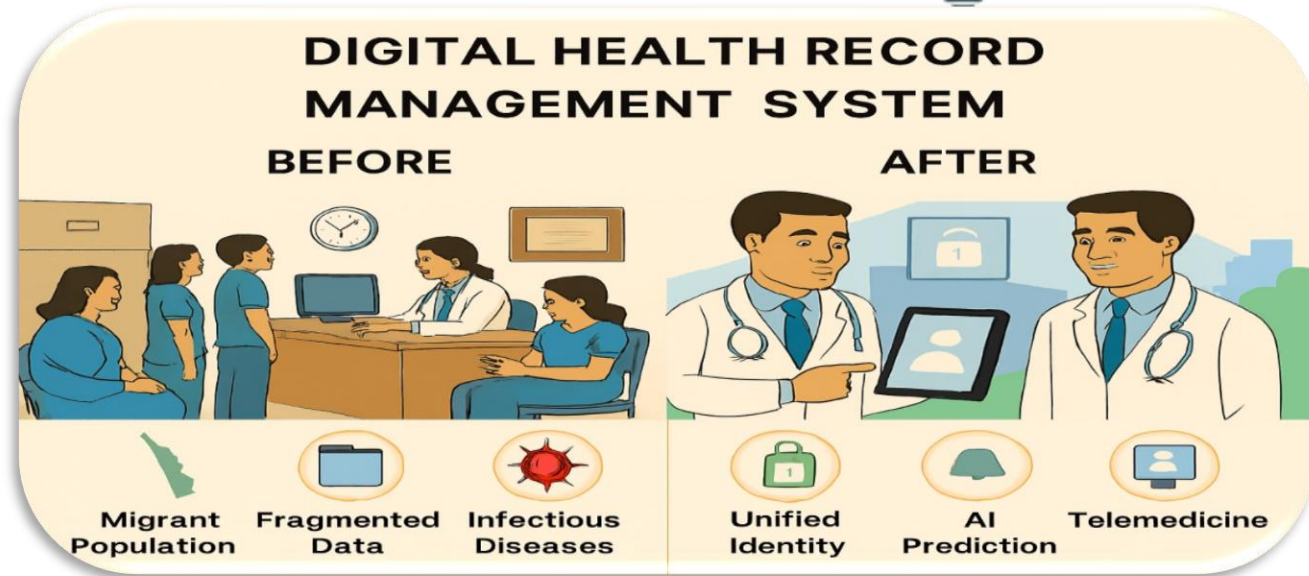
MIGRANT HEALTHCARE IN KERALA





Feasibility:

- **Tech Stack Ready** → Supports **Android, IOS, Windows, Mac, Linux and Web**.
- **Scalable Infrastructure** → **Cloud + Blockchain** for secure storage.
- **Workflow Defined** → **Hospital ↔ Doctor ↔ Patient** integration.
- **Interoperability** → Works across **hospitals & regions**.
- **Traditional Phone support** -> Via **SMS** chat and **IVR** support



Challenges:

- **Data Protection Risks** → **Privacy & cyber-security** issues.
- **Adoption Barriers** → **Resistance** from hospitals & workers.
- **Connectivity Gaps** → Rural **network** limitations.
- **AI Reliability** → Bias & prediction **accuracy** concerns.

Viability – Solutions:

- **Security** → End-to-end **encryption (AES-256) + blockchain (consortium - Hyperledger Fabric)** audit trails.
- **Adoption** → Awareness **campaigns & training** workshops.
- **Connectivity** → Offline mode with **auto-sync** support.
- **AI Governance** → Regular **audits & bias testing**.
- **Cost Sharing** → **Insurance** integration.

IMPACT AND BENEFITS

TARGET AUDIENCE



FASTER TREATMENT



ACCESSIBLE HEALTHCARE



EARLY OUTBREAK DETECTION



Potential Impact :

- **Target Audience** → Migrant Workers & Healthcare System
- **Accessible Healthcare** → Portable health records anytime, anywhere
- **Faster Treatment** → Doctors get complete patient history instantly
- **Early Outbreak Detection** → Protects both migrants & local communities

Benefits:

✓ Social

- Inclusive healthcare for all
- Better disease prevention
- Increased health awareness

✓ Economic

- Improved workforce productivity
- Operational efficiency through digital records

✓ Environmental

- Paperless records & digital prescriptions
- Eco-friendly operations

✓ Government

- Real-time health dashboards
- Data-driven policy decisions

IMPACT AND BENEFITS OF DHRMS FOR MIGRANT WORKERS IN KERALA

FOR MIGRANT WORKERS



- One portable health ID
- Access to affordable healthcare
- SOS alerts for emergencies

FOR GOVERNMENT

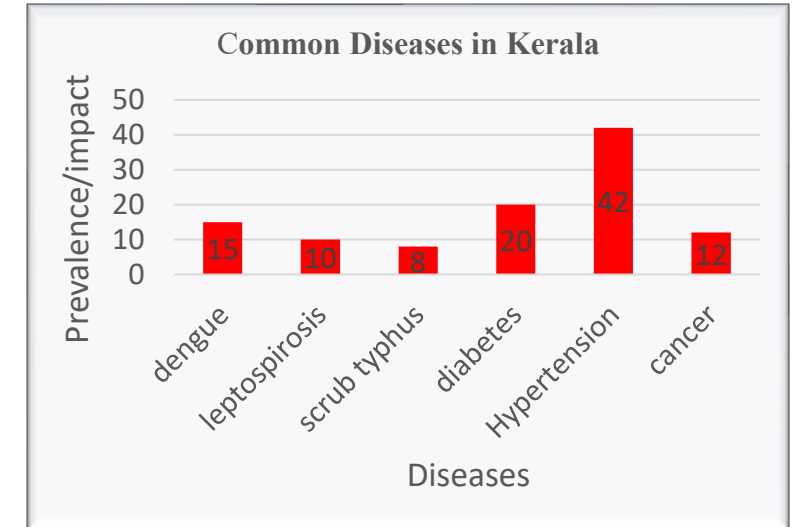
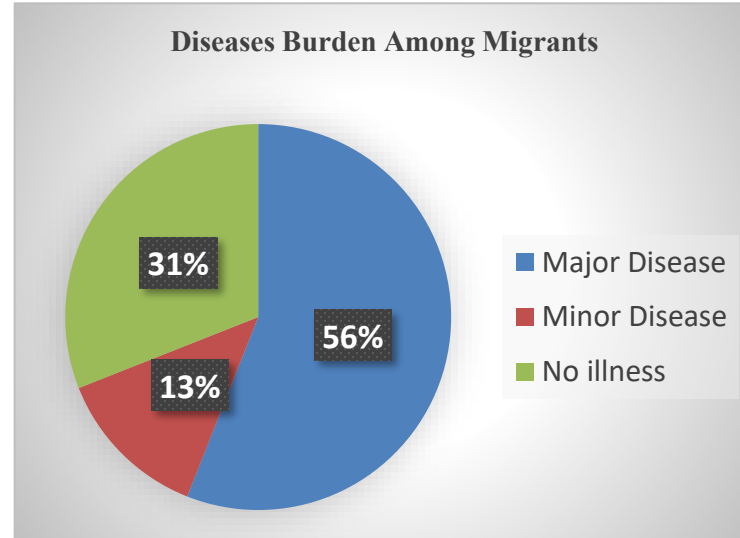
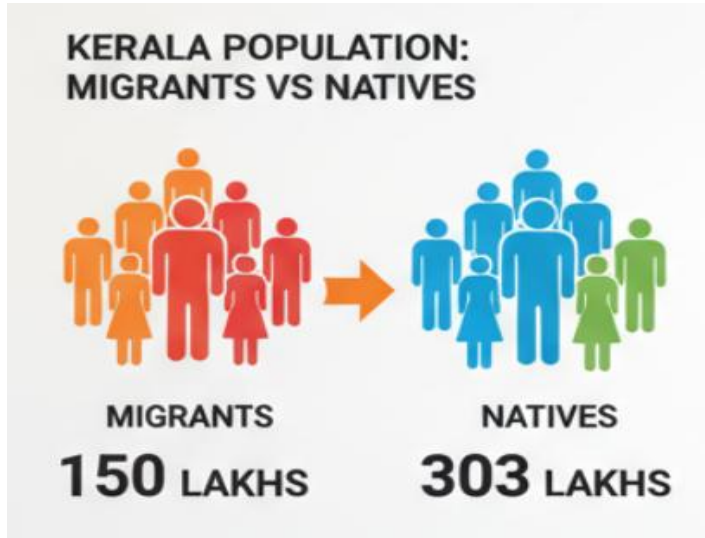


- Real-time outbreak monitoring
- Disease heatmaps for better resource allocation

FOR EMPLOYERS



- Healthy workforce
- Easy compliance with govt health regulations



1. [Social Work Intervention Model for Improving Health-Seeking Behaviour of Interstate Migrant Workers in Kerala](#)
2. [Seamless Electronic Medical Record for Health Management System in Emergency Patients](#)
3. [Medical Record Management System Using Blockchain Technology](#)
4. [Smart Health Monitoring System with Artificial Intelligence](#)